

| [ENGLISH](#) | [JAPANESE](#) |

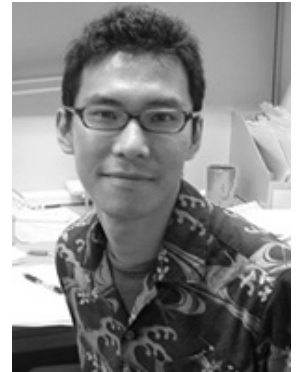
HIDEAKI SHIMAZAKI, Ph.D.

Program-specific Associate Professor at Kyoto University / Senior Scientist at Honda Research Institute Japan

Researcher at RIKEN BSI (2012-2016) [full CV](#)
Visiting researcher at RIKEN BSI / MIT (2008-2011)
JSPS Research Fellow (2006/4-2008/3, 2008/4-2011/3)

Ph.D. (Physics), 2007 Dept. of Physics, Kyoto University, Japan
M.A. (Neuroscience), 2004 Dept. of Neuroscience, Johns Hopkins University
B.E., 2000 Dept. of Applied Physics and Physico-Informatics, Keio University, Japan

Email: h.shimazaki@i.kyoto-u.ac.jp



[Histogram Optimization](#) | [Kernel Optimization](#) | [Estimating Time-varying Neural Interactions](#)
[Lecture notes](#) | [Group Photos](#) | [Lab webpage](#)

| [Research](#) | [Publications](#) | [Conferences](#) | [Awards](#) | [Curriculum Vitae](#) |

Updates

Recent publications

2020 Feb 12 Posted an arXiv paper by Aguilera and Moosavi [link](#)

A unifying framework for mean field theories of asymmetric kinetic Ising systems. *Miguel Aguilera, S. Amin Moosavi, Hideaki Shimazaki* [arXiv:2002.04309](#)

2020 Feb 27 Poster presentations at Cosyne2020 [link](#)

II-23 Inferring network motifs from neural activity using analytic input-output relation of LIF neurons. *Safura Rashid Shomali, Majid Nili Ahmadabadi, Seyyed Nader Rasuli, Hideaki Shimazaki*

III-80 Effects of structured neural correlations in population coding: beneficial or detrimental?. *Seyedamin Moosavi, Magalie Tatischeff, Bingyue Zhu, Hideaki Shimazaki*

2020年 Feb 14 Talk at Computational Principles in Active Perception and Reinforcement Learning in the Brain [link](#)

2020 Jan 28 Talk at Combining Information theoretic Perspectives on Agency [link](#)

2019 Dec 11 Updated a preprint on the complex ICA by MaBouDi, Krishna et al. [link](#)

2019 Dec 7 Talk at Singularity Salon at Tokyo (in Japanese) [link](#)

2019 Nov 16 Talk at Singuraity Salon at Osaka (in Japanese) [link](#)

2019 Nov 14 Uploaded a preprint on structured variational inference by spiking net by Shukla et al. [link](#)

2019 Sep 5 Edited Japanese Neural Network Society Journal

2019 Jul 18 **Jimmy Gaudreault received Best Student Award at IJCNN2019!** (2 students awards in 803 papers from 1532 submissions)

2019 Sep 29 Invited Talk Shimazaki The 7th International Congress on Cognitive Neurodynamics, Italy

2019 Sep 17 Talk by Miguel Aguilera et al., 15th Granada Seminar on Computational and Statistical Physics. [link](#)

2019. Sep 18 Poster by Safura Rashid Shomali et al., Bernstein Conference. Berlin, Germany [link](#)

2019 Aug-Sep We organized two events on Free energy principle at National Institute for Physiological Sciences. We invite Dr. Christopher Buckley at Sussex University as a lecturer.

Aug 31-Sep 1 **Tutorial workshop of the free energy principle of the brain**

Sep 2 Workshop **認知神経科学の先端 脳の理論から身体・世界へ**

2019 Aug 13 Invited Lecture Data science, Statistics & Visualization (DSSV2019), Kyoto

2019 Jun 7 Talk at Salon du Brain **Statistical nalysis of neural population activity: Toward thermodynamics of the brain** [link](#)

2019 Mar **Lecture and seminar at OIST** (Kenji Doya Lab) [link](#) [Lecture note](#)

2019 Mar We organized Systems Neuroscience Spring School: Statics and Dynamics of Neural Systems at Kyoto. [link](#)

2018 Nov Posted Dr. Shomali's paper on bioRxiv. [link](#)

2018 Jul I edited the special issue of Japanese Neural Network Society on the Free Energy Principle (in Japanese) [link](#)

2018 Jan **Neural Engine Hypothesis** was published as a chapter of Springer book. [link](#)

2017 Oct 5 An analysis paper on a LIF model by **Safura Shomali** et al. was accepted for publication in Journal of Computational Neuroscience.

2017 Sep 28 Special talk at Tamagawa University Brain Science Workshop: "Statistics of neural population activity".



2017 Jul 22 A co-authored paper published MaBouDi H, Shimazaki H, Giurfa M, Chittka L. **Olfactory learning without the mushroom bodies: Spiking neural network models of the honeybee lateral antennal lobe tract reveal its capacities in odour memory tasks of varied complexities.** PLoS

Computational Biology (2017) 13(6): e1005551. [\[link\]](#)

2017 Mar 3,4 Intensive lecture courses on the point process theory for "Society for Young Researchers in Neuroscience".



2017 Jan 13 A paper published. Donner, Obermeyer, Shimazaki, **Approximate Inference for Time-varying Interactions and Macroscopic Dynamics of Neural Populations**. PLoS Computational Biology (2017) 13(1): e1005309 [\[link\]](#)

2016 May 26 A co-authored paper was published. Mochizuki et al. **Similarity in Neuronal Firing Regimes across Mammalian Species**. *Journal of Neuroscience* (2016) 36:5736-5747. [\[link\]](#)

2016 Apr 1 A co-authored paper was published. Chou et al. **Social conflict resolution regulated by two dorsal habenular subregions in zebrafish**. *Science*, Vol. 352, Issue 6281, pp. 87-90 [\[link\]](#)

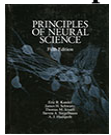
2015 Dec 24 I uploaded my recent thought on neural dynamcis. [Shimazaki H](#), **Neurons as an Information-theoretic Engine**. *arXiv:1512.07855* [\[pdf, link\]](#)

2015 May 15 A paper by HaDi MaBouDi was accepted for publication in Vision Research. MaBouDi H, Shimazaki H, Amari S, Soltanian-Zadeh H, "Representation of Higher-order Statistical Structures in Natural Scenes via Spatial Phase Distributions." [pdf](#)

2015 May Our optimization mehtod for kernel density estimation is being used in labs of leading neuroscientists. Hill, Fried & Koch [J Neurophysiol2015](#). Ito, Zhang, Witter, Moser & Moser [Nature 2015](#). Manita, Suzuki, ..., Larkum & Murayama [Neuron 2015](#).

2015 Mar 19 Our paper uncovering the role of silence in characterizing neural population activity has been accepted. Shimazaki, H., Sadeghi, K., Ishikawa, T., Ikegaya, Y. & Toyozumi, T. **Simultaneous silence organizes structured higher-order interactions in neural populations**. *Scientific Reports* 5, 9821 (2015) [\[link, preprint\]](#)

2014 April Japanese edition of *Principle of Neural Science* was published.



I translated the section, "Appendix F, Theoretical Approaches to Neuroscience: Examples from SingleNeurons to Networks" by Abbott, Fusi, and Miller. [\[link\]](#)

Lecture information

RIKEN Brain Science Training Program: Statistics for Neuroscience (Since 2014)

- [Link to RIKEN Brain Science Training Program.](#)
- [Link to a course website.](#)

2016 Feb **An introduction to the statistical analysis of point process networks** An extension lecture at Institute of Statistics and Mathematics with Prof. Shinsuke Koyama. [link](#)

2014, 2015 Winter, Statistical Physics at International Christian University

2013 Mar 3-12, Introduction to statistical models of neural spike train data
Two weeks lecture courses at the Institute for Research in Fundamental Sciences.

- [Link to a course website.](#)



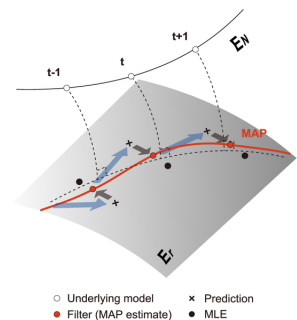
Research

| Research | Publications | Conferences | Awards | Curriculum Vitae |

Multiple Neurons: Dynamics of neural interactions and behavior

Classical neurophysiological studies are based on the idea that stimulus and behavioral information is encoded in the firing rates of single neurons. At the same time, precise spike coordination in the spikes of neuronal population is discussed as an indication of coordinated network activity relevant for information processing. Thus accurate assessment of neural correlations is a key to understand their coding principles.

Statistical structure of neural correlations may change while an animal is engaged in behavioral tasks. We suggest estimating the dynamics of neural correlations using a state-space method, and further developed a method to test their significance. With this method, we are investigating behavior-dependent interactions in neural populations.



A sample web application is available from here.

[Web Application]

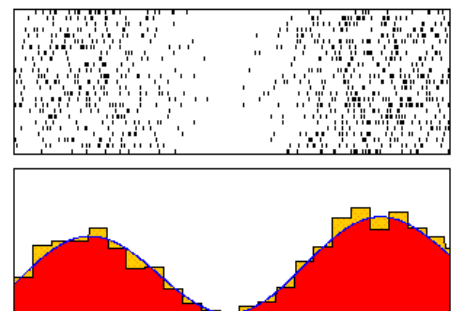
- Paper and proceedings
 - Gaudreault J, Saxena A, Shimazaki H. **Online estimation of multiple dynamic graphs in pattern**

- sequences.** (2019) IJCNN [arXiv:1901.07298](https://arxiv.org/abs/1901.07298) [pdf]
- Donner C., Obermeyer K., Shimazaki H. **Approximate Inference for Time-varying Interactions and Macroscopic Dynamics of Neural Populations.** PLoS Comp Biol (2017) 13(1): e1005309 [link, code]
 - Shimazaki H., Amari S., Brown E. N., and Gruen S., **State-space Analysis of Time-varying Higher-order Spike Correlation for Multiple Neural Spike Train Data.** PLoS Comput Biol 8(3): e1002385 (2012) [link](#) [PDF](#)
 - Shimazaki H., Single-trial estimation of stimulus and spike-history effects on time-varying ensemble spiking activity of multiple neurons: a simulation study. J. Phys.: Conf. Ser. (2013) [PDF](#)
 - Shimazaki H., Amari S., Brown E. N., and Gruen S. State-space Analysis on Time-varying Correlations in Parallel Spike Sequences, Proc. IEEE ICASSP2009, 3501-3504. [PDF](#)
 - Invited talks
 - ICSG2013. ELC International Meeting on "Inference, Computation, and Spin Glasses", Sapporo, Japan.
 - [NIPS2008 Workshop](#). "Statistical Analysis and Modeling of Response Dependencies in Neural Populations:", Whistler, Canada
 - [IEEE ICASSP2009](#). Special Session on 'Signal Processing for Neural Spike Trains', Taipei, Taiwan
 - Selected posters
 - Christian Donner and Hideaki Shimazaki. Approximation methods for inferring time-varying interactions of a large neural population. NIPS 2015 Workshop on 'Statistical Methods for Understanding Neural Systems'. Montreal, Canada. Dec 11, 2015
 - Neuro2014, Sharp, Shimazaki, Isomura and Fukai. Behaviour- and layer-dependent synchrony in motor cortex during volitional arm movement, Yokohama, Japan.
 - Cosyne2009. Shimazaki, Amari, Brown and Gruen. Detection of non-stationary higher-order spike correlation. [[abstract](#)]
 - Press
 - Press release in [English](#), [Japanese](#), [German](#).
 - [ScienseDaily](#), [REKEN RESEARCH Research Highlight](#)
 - Award
 - **Best student paper award at IJCNN2019** (Gaudreault J, Saxena A, Shimazaki H)
 - **Excellent paper award at ICONIP2016** (Donner C, Obermeyer K, and Shimazaki H)
 - **Japanese Neural Network Society Research Award 2009** to Shimazaki H, Amari S, Brown EN, and Gruen S Title: 'State-space Model of Dynamic Correlation'. JNNS2009.

Single Neurons: Optimizing Spike-rate Estimation

Neurons process sensory information using event sequences called action potentials or spikes. Much of the neurophysiological literature relies on the interpretation of the spike data by making a histogram or kernel density estimation of the spike-rate.

However, the quality of rate estimation is critically determined by the choice of bin-width or kernel bandwidth. We theoretically investigated accuracy and limitation in estimating the time-dependent spike-rate from finite samples, and developed methods to optimize the histogram bin-width and kernel bandwidth under objective error criteria.



Histogram Optimization

The optimal bin width Δ of a histogram is obtained as a minimizer of the formula,

$$C(\Delta) = \frac{2k - v}{\Delta^2}.$$

Here, k and v are mean and variance of the number of samples in the bins.

[\[Web Application\]](#)

- [1] Shimazaki and Shinomoto, A method for selecting the bin size of a time histogram, *Neural Computation* (2007) Vol. 19, No. 6, 1503-1527. [PDF](#)
- [2] Shimazaki and Shinomoto: A recipe for optimizing a time-histogram. *Advances in NIPS 2007 Vol. 19*, 1289-1296. [PDF](#)
- Award **Japanese Neural Network Society Young Researcher Award 2007**

Kernel Optimization

The optimal bandwidth w is obtained as a minimizer of the formula,

$$C(w) = \sum_{i,j} \int k_w(x - x_i) k_w(x - x_j) dx - 2 \sum_{i \neq j} k_w(x_i - x_j)$$

Here $k_w(t)$ and $\{x_i\}_{i=1}^N$ are a kernel function with bandwidth w and the data.

[\[Web Application\]](#)

- [3] Shimazaki and Shinomoto: Kernel Bandwidth Optimization in Spike Rate Estimation *Journal of Computational Neuroscience* (2010) Vol. 29 (1-2) 171-182. [PDF](#)
- [4] Shimazaki and Shinomoto: Kernel Width Optimization in the Spike-rate Estimation. *Neural Coding 2007* Abstract p120-123, Montevideo, Uruguay. [PDF](#)
- [5] Spike-rate Estimation with Locally Adaptive Kernel Method. *JNNS2008* p186-187 in Japanese.
- Award **Best Poster Award at Neural Coding 2007**

| Research | Publications | Conferences | Awards | Curriculum Vitae |

Publications

□ Preprint ■ Under Review ■ Published / In Press

[\[Google Scholar\]](#) [\[Researchmap\]](#) [\[ORCID\]](#) [\[Kyoto University DB\]](#)

Papers and Proceedings

- Miguel Aguilera, S. Amin Moosavi, [Hideaki Shimazaki](#). **A unifying framework for mean field theories of asymmetric kinetic Ising systems.** (2020) [arXiv:2002.04309](#)
- Shashwat Shukla, [Hideaki Shimazaki](#), Udayan Ganguly. **Structured mean-field variational inference and learning in winner-take-all spiking neural networks.** (2019) [arXiv:1911.05943](#) [\[pdf\]](#)
- [Hideaki Shimazaki](#). **The principles of adaptation in organisms and machines I: machine learning, information theory, and thermodynamics.** (2019) [arXiv:1902.11233](#) [\[pdf\]](#)
- Makio Torigoe, Tanvir Islam, Hisaya Kakinuma, Chi Chung Alan Fung, Takuya Isomura, Hideaki Shimazaki, Tazu Aoki, Tomoki Fukai, Hitoshi Okamoto. **Future state prediction errors guide active avoidance behavior by adult zebrafish.** (2019) doi: <https://doi.org/10.1101/546440> [\[pdf\]](#)
- Jimmy Gaudreault, Arunabh Saxena, Hideaki Shimazaki. **Online estimation of multiple dynamic graphs in pattern sequences.** (2019) IJCNN2019 [arXiv:1901.07298](#) [\[pdf\]](#)
- Safura Rashid Shomali, Majid Nili Ahmadabadi, Seyyed Nader Rasuli, Hideaki Shimazaki. **Uncovering network architecture using an exact statistical input-output relation of a neuron model.** (2018) bioRxiv. doi: <https://doi.org/10.1101/479956>
- Jimmy Gaudreault and Hideaki Shimazaki. **State-space analysis of an Ising model reveals contributions of pairwise interactions to sparseness, fluctuation, and stimulus coding of monkey V1 neurons.** *Lecture Notes in Computer Science* (2018) 11141, 641–651 [\[link\]](#)

- 島崎秀昭. 認識と行動の適応原理. 日本神経回路学会誌. 解説論文 In press (2018)
- Safura Rashid Shomali, Majid Nili Ahmadabadi, Hideaki Shimazaki, Seyyed Nader Rasuli. **How does transient signaling input affect the spike timing of postsynaptic neuron near the threshold regime: an analytical study.** *Journal of Computational Neuroscience* (2018) 44(2), 147-171 [[link](#), [pdf](#)]
- Kass RE, Amari S, Arai K, Diekmann CO, Diesmann M, Doiron B, Fairhall A, Fiddymont GM, Fukai T, Grün S, Harrison MT, Helias M, Nakahara H, Teramae J, Thomas PJ, Reimers M, Rodu J, Rotstein HG, Shea-Brown E, Shimazaki H, Shinomoto S, Yu BM, Kramer MA. **Computational Neuroscience: Mathematical and Statistical Perspectives.** *Annual Review of Statistics and Its Application.* in press [[link](#)]
- MaBouDi H, Shimazaki H, Giurfa M, Chittka L. **Olfactory learning without the mushroom bodies: Spiking neural network models of the honeybee lateral antennal lobe tract reveal its capacities in odour memory tasks of varied complexities.** *PLoS Computational Biology* (2017) 13(6): e1005551. [[link](#)]
- MaBouDi H, Subramani K, Soltanian-Zadeh H, Amari S, Shimazaki H. **Learning complex representations from spatial phase statistics of natural scenes.** *bioRxiv* (2017) doi: <https://doi.org/10.1101/112813>
- Donner C, Obermeyer K, Shimazaki H. **Approximate inference for time-varying interactions and macroscopic dynamics of neural populations.** *PLoS Computational Biology* (2017) 13(1): e1005309 [[link](#), [code](#)]
- Donner C and Shimazaki H. **Approximate inference method for dynamic interactions in larger neural populations.** *The 23rd International Conference on Neural Information Processing (ICONIP 2016) LNCS* (2016) 9949, 104–110
- Mochizuki Y, Onaga T, Shimazaki H, Shimokawa T, Tsubo Y, Kimura R, Saiki A, Sakai Y, Isomura Y, Fujisawa S, Shibata K, Hirai D, Furuta T, Kaneko T, Takahashi S, Nakazono T, Ishino S, Sakurai Y, Kitsukawa T, Lee JW, Lee H, Jung MW, Babul C, Maldonado PE, Takahashi K, Arce-McShane FI, Ross CF, Sessle BJ, Hatsopoulos NG, Brochier T, Riehle A, Chorley P, Gruen S, Nishijo H, Ichihara-Takeda S, Funahashi S, Shima K, Mushiake H, Yamane Y, Tamura H, Fujita I, Inaba N, Kawano K, Kurkin S, Fukushima K, Kurata K, Taira M, Tsutsui K, Ogawa T, Komatsu H, Koida K, Toyama K, Richmond BJ, and Shinomoto S. **Similarity in Neuronal firing regimes across mammalian species.** *Journal of Neuroscience* (2016) 36:5736-5747. [[link](#)]
- Chou M.Y., Amo R., Kinoshita M., Chong B.W., Shimazaki H, Agetsuma M., Shiraki T., Aoki T., Takahoko M., Yamazaki M., Higashijima S., and Okamoto H. **Social conflict resolution regulated by two dorsal habenular subregions in zebrafish.** *Science*, (2016) Vol. 352(6281) 87-90 [[link](#)]
- MaBouDi H, Shimazaki H, Amari S, Soltanian-Zadeh H, **Representation of higher-order statistical structures in natural scenes via spatial phase distributions.** *Vision Research* (2016) 120, 61–73 [[pdf](#), [link](#)]
- Shimazaki H, **Neurons as an Information-theoretic Engine.** *arXiv*:1512.07855 (2015) [[pdf](#), [link](#)]
- Shimazaki H, Sadeghi K, Ishikawa T, Ikegaya Y, Toyoizumi T, **Simultaneous silence organizes structured higher-order interactions in neural populations.** *Scientific Reports* (2015) 5, 9821. [[link](#)]
- Shimazaki H, **Single-trial estimation of stimulus and spike-history effects on time-varying ensemble spiking activity of multiple neurons: a simulation study.** *J. Phys.: Conf. Ser.* (2013) 473 012009. [[link](#), [pdf](#)]
- Shimazaki H, Amari S, Brown EN, and Gruen S, **State-space analysis of time-varying higher-order spike correlation for multiple neural spike train data.** *PLOS Computational Biology* (2012) 8(3): e1002385. [[link](#), [pdf](#), [support page](#)]
- Shimazaki H, **Analysis of multiple neural spike train data using the log-linear model : From stationary to time-varying spike correlation.** *The Brain & neural networks* (2011) 8(4) 194-203 (In Japanese) [[link](#), [pdf](#)]
- Shimazaki H and Shinomoto S, **Kernel bandwidth optimization in spike rate estimation.** *Journal of Computational Neuroscience* (2010) Vol. 29 (1-2) 171-182 [[link](#), [pdf](#), [support page](#)]
- Shimazaki H, Amari S, Brown EN, and Gruen S, **State-space analysis on time-varying correlations in parallel spike sequences.** *Proc. IEEE ICASSP2009* (2009) 3501-3504 [[link](#)]

- [Shimazaki H](#) and Shinomoto S, **A method for selecting the bin size of a time histogram**, *Neural Computation* (2007) Vol. 19(6), 1503-1527 [[link](#), [pdf](#), [support page](#)]
- [Shimazaki H](#) and Shinomoto S, **A recipe for optimizing a time-histogram**, *Advances in Neural Information Processing Systems* (2007) Vol. 19, 1289-1296 [[pdf](#), [support page](#)]
- [Shimazaki H](#) and Niebur E, **Correlated multiplicative modulation in coupled oscillator systems: a model of selective attention**, *Progress of Theoretical Physics Supplement* (2006) No.161 336-339 [[link](#)]
- [Shimazaki H](#) and Niebur E, **Phase transitions in multiplicative competitive processes**, *Physical Review E* (2005) 72(1), 011912 [[link](#), [pdf](#)]
- Tsukada M, Aihara T, Kobayashi Y, [Shimazaki H.](#), **Spatial analysis of spike-timing-dependent LTP and LTD in the CA1 area of hippocampal slices using optical imaging**, *Hippocampus* (2005) 15(1), 104-109. [[link](#)]
- Kobayashi Y, [Shimazaki H.](#), Aihara T, Tsukada M. **Spatial distributions of hippocampal LTP/LTD induced electrically from schaffer collaterals and stratum oriens with relative timing**, *The Brain & neural networks* (2001) Vol.8, No.2 p.57-64. (In Japanese) [[link](#)]

Book chapter

- Neural Engine Hypothesis. In: Chen Z., Sarma S. (eds) *Dynamic Neuroscience*. (2018) Springer, Cham [link](#)

Translation

- Principles of Neural Science 5th edition, Kandel et al.; McGraw-Hill Professional (2012) "Appendix F, Theoretical Approaches to Neuroscience: Examples from Single Neurons to Networks" by Abbott, Fusi, and Miller. [[link](#)]

| Research | Publications | Conferences | Awards | Curriculum Vitae |

Conference, Workshop, Seminar (Those written in Japanese were done in Japanese.)

■ 2020

Seyedamin Moosavi, Magalie Tatischeff, Bingyue Zhu, Hideaki Shimazaki. Effects of structured neural correlations in population coding: beneficial or detrimental? [Computational and Systems Neuroscience \(Cosyne\) 2020](#). Feb 27 - Mar 1. Denver, USA **POSTER**

Hideaki Shimazaki. The brain as an information-theoretic engine: A new paradigm for quantifying perceptual capacity of neural dynamics. Combining Information theoretic Perspectives on Agency. Jan 28-29, The University of Tokyo, Japan. **TALK** [link](#)

■ 2019

島崎 秀昭. 脳の理論の過去・未来：ベイズ脳仮説からニューラルエンジンへ。 シンギュラリティサロン東京, 大手町サンケイプラザ 2019年12月7日 **TALK** [link](#)

島崎 秀昭. 脳の理論の過去・未来：ベイズ脳仮説からニューラルエンジンへ シンギュラリティサロン大阪, グランフロント大阪・ナレッジサロン・プレゼンラウンジ 2019年11月16日 **TALK** [link](#)

Hideaki Shimazaki*. Dynamic neural interactions revealed by the state-space Ising model. The 7th International Congress on Cognitive Neurodynamics. Sep 29. Alghero, Italy. **TALK** [link](#)

Safura Rashid Shomali*, Majid Nili Ahmadabadi, Seyyed Nader Rasuli, Hideaki Shimazaki
Judging between Excitation and Inhibition: Identifying Local Network Architecture by an Analytic Pre-Post Relation.

Bernstein Conference 2019. Sep 18, Berlin, Germany. **POSTER** [link](#)

Miguel Aguilera*, Amin Seyed Moosavi, Hideaki Shimazaki. Unifying framework for mean field theories of asymmetric kinetic Ising systems . 15th Granada Seminar on Computational and Statistical Physics: Stochastic and Collective Effects in Neural Systems. Sep 17. Universidad de Granada, Spain. **TALK** [link](#)

島崎 秀昭. 学習と認識の熱力学：ニューラルエンジンとはなにか？ 生理研研究会2019 認知神経科学の先端「脳の理論から身体・世界へ」 岡崎 2019年9月2日 **TALK**

Hideaki Shimazaki. Visualizing dynamics of cooperative activities of neurons for neural coding studies. Data Science, Statistics, & Visualisation (DSSV2019) Aug. 13. Doshisha University, Kyoto, Japan. **TALK** [link](#)

島崎 秀昭. 神経細胞集団活動の統計解析：脳の熱力学に向けて. サロン・ド・脳. 京都大学 2019年6月7日 **TALK**

Hideaki Shimazaki. Network architecture underlying sparse neural activity characterized by structured higher-order interactions. Okinawa Institute of Technology (OIST). Okinawa, Japan. Mar 26 2019 **TALK** [link](#)

S. Amin Moosavi, Hideaki Shimazaki. Role of gain-control and neural correlations in efficient stimulus coding. Consciousness Research Network (CoRN 2019) Okazaki, Japan. Feb 23 2019 **POSTER**

Magalie Tatischeff, Jimmy Gaudreault, Christian Donner, Hideaki Shimazaki. Thermodynamic analysis of neural populations by the state-space Ising model. Consciousness Research Network (CoRN 2019). Okazaki, Japan. Feb 23 2019 **POSTER**

■ 2018

Hideaki Shimazaki, Magalie Tatischeff, Jimmy Gaudreault, Christian Donner. Thermodynamic analyses of neural populations. Analysis and Synthesis for Human/Artificial Cognition and Behaviour, A satellite workshop of JNNS2018. Okinawa Institute of Science and Technology Graduate University. Oct 22, 2018 **POSTER** [link](#)

Jimmy Gaudreault and Hideaki Shimazaki. State-space analysis of an Ising model reveals contributions of pairwise interactions to sparseness, fluctuation, and stimulus coding of monkey V1 neurons. The 27th International Conference on Artificial Neural Networks (ICANN2018) Oct 5, 2018 **POSTER**

島崎秀昭 スパースな集団活動を生み出す神経ネットワーク構造. 大規模ネットワーク上のイベント生起ダイナミクスの予測と制御 軽井沢 2018年7月6日 **TALK** [link](#)

Jimmy Gaudreault, Arunabh Saxena, Hideaki Shimazaki. State-space analysis of an Ising model reveals contributions of pairwise interactions to sparseness, fluctuation, and stimulus coding of monkey V1 neurons. 脳と心のメカニズム第17回冬のワークショップ 北海道 2018年1月9日 **POSTER**

■ Seminar, Conference, Workshop in 2017

Shimazaki H. Network architectures underlying variable, sparse population activity of neurons. Fluctuations of event occurrences in a variety of networks. Kyoto, Japan. Nov 15, 2017 **TALK** [link](#)

島崎秀昭 神経細胞集団活動の統計数理 玉川大学脳科学ワークショップ 石和温泉 2017年9月28日 **TALK (Invited)**

Makio Torigoe, Islam Tanvir, Hisaya Kakinuma, Hideaki Shimazaki, Chi Chung Alan Fung, Tazu Aoki, Tomoki Fukai, Hitoshi Okamoto In-vivo imaging of telencephalic neural activities in adult zebrafish performing decision making task in the closed-loop virtual reality environment. Annual Meeting of the Japan Neuroscience Society (Neuro2017). Makuhari Messe, Japan. Jul 23 2017. **TALK by Torigoe**

島崎秀昭 神経活動データの時系列モデリング入門 第8回脳科学若手の会合宿 ホテルウィングインターナシヨ

ナル相模原 2017年3月4, 5日 **Lecture (Invited)**

■ 2017

Shimazaki H. Network architectures underlying variable, sparse population activity of neurons. Fluctuations of event occurrences in a variety of networks. Kyoto, Japan. Nov 15, 2017 **TALK** [link](#)

島崎秀昭 神経細胞集団活動の統計数理 玉川大学脳科学ワークショップ 石和温泉 2017年9月28日 **TALK (Invited)**

Makio Torigoe, Islam Tanvir, Hisaya Kakinuma, Hideaki Shimazaki, Chi Chung Alan Fung, Tazu Aoki, Tomoki Fukai, Hitoshi Okamoto In-vivo imaging of telencephalic neural activities in adult zebrafish performing decision making task in the closed-loop virtual reality environment. Annual Meeting of the Japan Neuroscience Society (Neuro2017). Makuhari Messe, Japan. Jul 23 2017. **TALK by Torigoe**

島崎秀昭 神経活動データの時系列モデリング入門 第8回脳科学若手の会合宿 ホテルウィングインターナショナル相模原 2017年3月4, 5日 **Lecture (Invited)**

■ Seminar, Conference, Workshop in 2016

Donner C. and Shimazaki H., Estimating dynamic functional networks of larger neural populations. Society for Neuroscience 2016. Nov 16, 2016 **POSTER**

Shomali S.R., Ahmadabadi M.N., Rasuli N. S., Shimazaki H., Exact analysis of spiketiming and higherorder interactions of neurons at the threshold regime suggests network architecture underlying sparse population activity. Society for Neuroscience 2016. Nov 14, 2016 **POSTER**

Donner C. and Shimazaki H., Large-scale inference of time-varying neural interactions. The 23rd International Conference on Neural Information Processing (ICONIP 2016). Kyoto, Japan. Oct 16, 2016 (Talk by Donner) **EXCELLENT PAPER AWARD**

MaBouDi H., Shimazaki H., Lars Chittka L. Modelling elemental learning of honeybees by spiking neural networks. EURBEE 2016. Cluj-Napoca, Romania. Sep 7, 2016 (Talk by MaBouDi)

Shimazaki H. Toward thermodynamic principles of consciousness. University of Sussex, UK. Jul 7, 2016. **TALK**

Shimazaki H. Higher-order interactions of neural populations. University College London, UK. Jul 5, 2016. **TALK**

Shimazaki H. Population coding of neurons: Dynamics, higher-order interactions, and mechanisms. Queen Mary University of London, UK. Jul 4, 2016. **TALK**

Shomali S.R., Ahmadabad M.N., Shimazaki H., Rasuli SN. Exact spike-timing distribution reveals higher-order interactions. CNS*2016. Jeju, South Korea. Jul 2, 2016 (Poster by Shomali) **POSTER (Reviewed)** [links](#)

島崎秀昭, Andrew J Peters, 小宮山尚樹, 豊泉太郎. Redundant spatiotemporal coding by layer 2/3 neurons of motor cortex during initial motor learning. 第17回ノンパラメトリック統計解析とベイズ統計 慶應義塾大学三田キャンパス 2016年3月29・30日 **TALK**

島崎秀昭 神経ネットワークの情報コーディング：ダイナミクス・高次相関・メカニズム. 第4回東工大若手物性セミナー 東京工業大学すずかけ台キャンパス 2016年2月18日 **TALK (Invited)**

島崎秀昭 動的イジングモデルを用いた神経回路網活動の解析 NS Forum 国際基督教大学 2016年2月2日 **TALK (Invited)**

島崎秀昭, Andrew J Peters, 小宮山尚樹, 豊泉太郎 Redundant coding by layer 2/3 neurons of motor cortex during initial motor learning. 脳と心のメカニズム 第16回冬のワークショップ 北海道ルスツ 2016年1月6,7,8日 **POSTER** [[link](#)]

■ *Seminar, Conference, Workshop in 2015*

Christian Donner and Hideaki Shimazaki. Approximation methods for inferring time-varying interactions of a large neural population. NIPS 2015 Workshop on 'Statistical Methods for Understanding Neural Systems'. Montreal, Canada. Dec 11, 2015 **POSTER** [[link1](#), [link2](#)]

島崎秀昭 神経回路網活動の熱力学的考察 数理モデリング研究会 滋賀 2015年11月28,29日 **TALK** [[link](#)]

Shimazaki H. Simultaneous silence explains structured higher-order interactions of neural populations. Juelich Research Center. Juelich, Germany. May 5, 2015 Seminar **TALK** [[link](#)]

Shimazaki H. Simultaneous silence explains structured higher-order interactions of neural populations. Bernstein Center of Computational Neuroscience Berlin. Berlin, Germany. Apr 28, 2015 **TALK** [[link](#)]

島崎秀昭 Simultaneous silence explains structured higher-order interactions of neural populations. 第16回ノンパラメトリック統計解析とベイズ統計 慶應義塾大学 2015年3月25日 **TALK**

島崎秀昭 神経細胞の集団活動の高次統計とメカニズム 第3回ヘテロ・ニューロ・アナリシス研究会 玉川大学 2015年3月18日 **TALK**

■ *Seminar, Conference, Workshop in 2014*

Thomas Sharp, Hideaki Shimazaki, Yoshikazu Isomura and Tomoki Fukai. State-space analysis of behaviour- and layer-dependent synchrony in motor cortex during volitional arm movement. Society for Neuroscience (SfN) 2014, Washinton DC, USA. Nov 16, 2014 **POSTER**

Thomas Sharp, Hideaki Shimazaki, Yoshikazu Isomura and Tomoki Fukai. Behaviour- and layer-dependent synchrony in motor cortex during volitional arm movement, Neuro2014, Yokohama, Japan. Sep 11, 2014 **POSTER**

Safura Rashid Shomali, Majid Nili Ahmadabadi, Hideaki Shimazaki, S Nader Rasuli. Theoretical study on spike-timing probability in a pair of pre-post synaptic neurons. Neuro2014, Yokohama, Japan. Sep 11, 2014 **POSTER**

Shimazaki H. Simultaneous silence explains structured higher-order interactions of neural populations. June 3 11:00-12:00 Place: CiNet 1F Conference Room A, **TALK**

Shimazaki H. Simultaneous silence explains structured higher-order interactions of neural populations. 京都大学セミナー, June 2 **TALK**

Tom Sharp*, Hideaki Shimazaki, Yoshikazu Isomura and Tomoki Fukai. Behaviour- and layer-dependent synchrony in motor cortex during volitional arm movement. Workshop on data mining in neuroscience. National Institute of Informatics, Tokyo, Japan. May 28-29, 2014 **TALK**

HaDi MaBouDi, Hideaki Shimazaki, Hamid Soltanian-Zadeh, Shun-ichi Amari. Bimodal Distributions of Local Phase Variables in Natural Images. The 2014 VSS Annual Meeting, St. Pete Beach, Florida. May 18, 2014 **POSTER**

HaDi MaBouDi, Hideaki Shimazaki, Mehdi Abouzari, Shun-ichi Amari, Hamid Soltanian-Zadeh, Statistical inference for directed phase coupling in neural oscillators. [Computational and Systems Neuroscience \(Cosyne\) 2014](#). Feb 27 Salt Lake City, USA. **POSTER**

島崎秀昭. State-space analysis of higher-order interactions in parallel event sequences 第15回ノンパラメトリック統計解析とベイズ統計 慶應義塾大学三田キャンパス 2014年3月19日 **TALK**

島崎秀昭. 高次相関を伴う神経回路網活動：局所回路の計算原理を求めて. 神経科学と統計科学の対話 4. 統計数理研究所 2013年3月17日 TALK

Shimazaki H., Sadeghi K., Ikegaya Y., Toyozumi T. Structured higher-order interactions explain the simultaneous silence of neural populations. [脳と心のメカニズム 第14回 冬のワークショップ](#) 2014年1月8日(水)-10日(金) ポスター発表

島崎秀昭「留学という選択」 大学院留学・ポスドク留学・語学留学・サマースクールなど異なるタイプの留学についてお話しします。また日本に来る留学生への対応や海外での講義体験などの国際交流についてもご紹介いたします。 [脳科学若手の会](#) 2014年1月25日(日)

■ Seminar, Conference, Workshop in 2013

Shimazaki H., Sadeghi K., Ikegaya Y., Toyozumi T., [The simultaneous silence of neurons explains structured higher-order interactions in ensemble spiking activity](#), SfN2013. **POSTER**

Shimazaki H., TBA, ELC International Meeting on "Inference, Computation, and Spin Glasses" (ICSG2013), Sapporo, Japan. July 28-30, 2013 **TALK (Invited)**

Shimazaki H., Higher-order interactions in population activity of hippocampal CA3 neurons, Workshop on statistical analysis of neurophysiological and clinical data, Kyoto, Japan. **TALK**

Shimazaki H., Estimating Time-varying Higher-order Neuronal Interactions in Awake Behaving Animals, [Modeling Neural Activity: Statistics, Dynamical Systems, and Networks](#), Lihue, Hawaii, USA. 2013 June 26-28 **TALK (Selected)**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyozumi T., The simultaneous silence of neurons explains higher-order interactions in ensemble spiking activity, Neuro2013, Kyoto, Japan. Jun. 20 **POSTER**

Shimazaki H. Estimating dynamic neural interactions in awake behaving animals. [The 3rd Mathematical Neuroscience Workshop](#) in School of Mathematics, Institute for Research in Fundamental Sciences (IPM). Tehran, Iran. Mar. 13 **TALK (Invited)**

Shimazaki H. The Simultaneous Silence of Neurons Explains Higher-Order Interactions in Ensemble Spiking Activity. [The 3rd Mathematical Neuroscience Workshop](#) in School of Mathematics, Institute for Research in Fundamental Sciences (IPM). Tehran, Iran. Mar. 13 **TALK (Invited)**

島崎秀昭. 高次相関を伴う神経回路網の活動と情報コーディング [玉川大学脳科学若手の会](#) 第88回談話会 Feb 22 **TALK (Invited)**

Shimazaki H., Sadeghi K., Ikegaya Y., Toyozumi T. The simultaneous silence of neurons explains structured higher-order interactions in spontaneous spiking activity. [神経科学と統計科学の対話 3](#). 統計数理研究所 Feb 18-19 **TALK**

■ Seminar, Conference, Workshop in 2012

Shimazaki H., Kolia Sadeghi, Yuji Ikegaya, Taro Toyozumi. Joint inactivation statistics of population spiking activities. RIKEN BSI Retreat 2012, Nov 12-13, Karuizawa, Japan

Shimazaki H. The simultaneous silence of neurons explains structured higher-order interactions in ensemble spiking activity. BSI Lunch Seminar 2012 Nov. 8

Shimazaki H., Kolia Sadeghi, Yuji Ikegaya, Taro Toyozumi. Joint inactivation statistics of population spiking activities. [Workshop on statistical aspects of neural coding](#). Nov 1-2. Kyoto University & Ritsumeikan University.

Shimazaki H. Tracking Dynamic Neural Interactions in Awake Behaving Animals. [Workshop on neural information](#)

[flow](#), Kyoto University, Kyoto Jun 20 2012 **TALK (Invited)**

島崎秀昭, 篠本滋. ヒストグラム・カーネル密度推定の神経スパイクデータへの適用：理論と実践 第13回ノンパラメトリック統計解析とベイズ統計 2012年3月29・30日 慶應義塾大学三田キャンパス **TALK (Invited)**

Shimazaki H., Kolia Sadeghi, Yuji Ikegaya, Taro Toyoizumi, The simultaneous silence of neurons explains higher-order interactions in ensemble spiking activity. [Computational and Systems Neuroscience \(Cosyne\) 2012](#). Salt Lake City, USA. 2012 Feb 23-26 **POSTER (Reviewed)**

Shimazaki H. 非線形セミナー 神経細胞の高次スパイク相関：状態空間モデルによる解析. 京都大学理学部物理学第一教室 2012 Feb 2 **TALK (Invited)**

■ *Seminar, Conference, Workshop in 2011*

島崎秀昭, 小山慎介. 神経スパイク解析における状態空間モデル, GLMの応用. 統計科学と神経科学の対話 2011年 12月26-27日 [**TALK, Invited**]

Shimazaki H., Amari S., Brown E. N., and Gruen S. Dynamics of Higher-order Spike Correlation in an Awake Behaving Monkey: Analysis by a State-space Model. RIKEN BSI Retreat, Karuizawa, Japan. Oct 31 **POSTER**

Shimazaki H., Ikegaya Y., and Toyoizumi T. A New Sparse Log-linear Model for Simultaneously Active and Inactive Neurons. RIKEN BSI Retreat, Karuizawa, Japan. Oct 31 **POSTER**

Shimazaki H. and Brown E.N. Copula-based Mixture Time-series Model of Continuous and Point Processes for Synthetic Analysis of Neural Signals. RIKEN BSI Retreat, Karuizawa, Japan. Oct 31 **POSTER**

島崎秀昭 脳の高次機能と動的高次スパイク相関：状態空間モデルによる解析, [統計神経科学ミニワークショップ](#) 統計数理研究所 9月8日 [**TALK, Invited**]

Shimazaki H. and Brown E. N. Constructing a joint time-series model of continuous and Bernoulli/Poisson processes using a copula [Computational and Systems Neuroscience 2011](#) Salt Lake City, USA. Feb 24-27 **POSTER (Reviewed)**

■ *Seminar, Conference, Workshop in 2010*

Shimazaki H. Analysis of Dynamic Neural Spike Data: From Firing Rates to Spike Correlations. Neurostatistics Working Group Seminar, Dept. of Biostatistics, Harvard University, Dec 1 [**TALK, Invited**] [link](#)

Shimazaki H. Detection of dynamic cell assemblies by the Bayes Factor. [Workshop on spatio-temporal neuronal computation](#), Kyoto University, Japan Sep 6-7 [**TALK, Invited**]

Shinomoto S*, Shimazaki H., and Shimokawa T. Characterizing neuronal firing with the rate and the irregularity. Neuro2010, Kobe, Japan Sep. 2 S3-10-1-3 [[abst](#)]

Shimazaki H., Gruen S., and Amari S. Analysis of subsets of higher-order correlated neurons based on marginal correlation coordinates. [Cosyne 2010, Salt Lake City, USA](#). 1-63 **Feb 25-28** [[abst](#)]

■ *Seminar, Conference, Workshop in 2009*

Shimazaki H., Amari S., Brown E. N., and Gruen S. State-space Model of Dynamic Spike Correlation. [Japanese Neural Network Society 2009](#). Sendai, Japan. Sept. 24-26. **Talk+Poster** [[Japanese Neural Network Society 2009 Distinguished Research Award](#)]

Shimazaki H. Analysis of Dynamic Spike Data: From Spike Rate to Multiple Neuron Spike Correlation. The 6th Brain

Lunch Seminar. RIKEN Brain Science Institute, Saitama, Japan. Sep. 8. TALK

Shimazaki H., Amari S., Brown E. N., and Gruen S. Bayes Factor Analysis for Detection of Time-dependent Higher-order Spike Correlations. [CNS2009 in Berlin, Germany](#). Jul 18-23 P99 POSTER [[abstract](#)]

Shimazaki H. and Shinomoto S. Histogram binwidth and kernel bandwidth selection for the Spike-rate estimation. [CNS2009, Berlin, Germany](#). Jul 18-23 P116 POSTER [[abstract](#)]

Shimazaki H., Amari S., Brown E. N., and Gruen S. Estimating time-varying spike correlations from parallel spike sequences, German-Japanese Workshop "Computational and Systems Neuroscience", Berlin, May 25-28, 2009 POSTER

Shimazaki H., Amari S., Brown E. N., and Gruen S. State-space Analysis on Time-varying Correlations in Parallel Spike Sequences. IEEE ICASSP2009 Special Session on 'Signal Processing for Neural Spike Trains', Taipei, Taiwan. Apr 24 SS-L10.4 [**LECTURE (Invited)**]

Shimazaki H., Amari S., Brown E. N., and Gruen S. Detection of non-stationary higher-order spike correlation. [Cosyne 2009](#), Salt Lake City, USA. Feb 26 -Mar 3 POSTER II-62 [[abstract](#)]

■ Seminar, Conference, Workshop in 2008

Shimazaki H., Amari S., Brown E. N., and Gruen S. State-space Analysis on Time-varying Higher-order Spike Correlations. NIPS2008 Workshop on 'Statistical Analysis and Modeling of Response Dependencies in Neural Populations', Whistler, Canada Dec 13, 17:00-17:30 [[abstract](#)] [**TALK (Invited)**]

Shimazaki H. and Gruen S. Selecting a state-space model of higher-order correlations in parallel spike trains from competing hierarchical log-linear models. [RIKEN BSI Retreat 2008](#), Karuizawa, Japan. Nov 4-5 POSTER

Shimazaki H. and Gruen S. [IBIS](#), Sendai, Japan. Oct 29 Poster. in Japanese.

Shimazaki H. and Shinomoto S. Spike-rate Estimation with Locally Adaptive Kernel Method . [Japanese Neural Network Society 2008](#). Sep 24-26 Tsukuba, Japan. PS3-4, p186-187 [PDF](#) in Japanese. [**Spotlight Poster**]

Shimazaki H. Smoothing methods for spike data: From single neurons rate estimation to multiple neurons correlation analysis. Joint Journal Club at RIKEN BSI. Jun 17

Shimazaki H. and Gruen S. Estimating time-dependent higher-order interactions in parallel spike trains. [Neuro2008](#), Tokyo, Japan. Jul 9-11 POSTER P1-w04 [[abstract](#)]

Shimazaki H., Brown E. N. and Gruen S. State-space Analysis on Time-dependent Correlation in Parallel Spike Trains. [Statistical Analysis of Neuronal Data \(SAND4\)](#), Pittsburgh, PA, USA. May 29-31 POSTER [[abstract](#)]

■ Seminar, Conference, Workshop in 2007

Shimazaki H. and Gruen S. Estimation of Time-dependent Higher-Order Interactions in Parallel Spike Trains. [Riken BSI Retreat](#), Karuizawa, Japan. Nov 26-28 POSTER

Shimazaki H. and Shinomoto S. Kernel Width Optimization in the Spike-rate Estimation. [Neural Coding 2007](#), Montevideo, Uruguay. Nov 7 POSTER [[pdf](#)] [**Best Poster Award**]

Shimazaki H. and Shinomoto S. Optimization of a Histogram of Spike Data. [Neuro2007](#), Yokohama, Japan. Sep 11 POSTER P2-k22

Shimazaki H. A Recipe for Optimizing a Time Histogram of Spike Data. [Riken BSI Forums](#) at RIKEN Brain Science Institute, Wako, Japan. Host: Sonja Gruen Apr 17 TALK

Shimazaki H. A recipe for constructing a Peri-stimulus Time Histogram. [The Boadian Seminar](#) at Mind/Brain Institute, Johns Hopkins University. Host: Ernst Niebur. Mar 1 TALK

Shimazaki H. and Shinomoto S. A recipe for optimizing a time-histogram with variable bin sizes. [Computational and Systems Neuroscience 2007](#), Salt Lake City, Feb 22-27 POSTER

▪ *Seminar, Conference, Workshop in 2006*

Shimazaki H. and Shinomoto S. A recipe for optimizing a time-histogram [Neural Information Processing Systems](#) Dec 4-9, Whistler, B. C. POSTER [[pdf](#)] [[Spotlight Poster](#)]

Shimazaki H. A method to optimize a time histogram by correcting onset latencies of spike sequences [Japanese Neural Network Society JNNS2006](#) Sep 1-21 Nagoya, Japan O4-1 ORAL + POSTER [[Japanese Neural Network Society 2007 Young Researcher Award](#)]

Shimazaki H. Self-organized criticality by natural selection. [Frontiers in Dynamics: Physical and Biological Systems](#) Tokyo, Japan, May 22-24, 2006, Poster [[abstract](#)]

Shimazaki H. and Shinomoto S. Recipes for constructing an optimal time histogram. [Statistical Analysis of Neuronal Data \(SAND3\)](#), Pittsburgh, PA, May 12-13, 2006, Poster [[abstract](#)]

| [Research](#) | [Publications](#) | [Conferences](#) | [Awards](#) | [Curriculum Vitae](#) |

Award, Grant & Financial Aid

Award

- **ICONIP2016 Excellent paper award**
- **Japanese Neural Network Society 2009 Research Award**
- **Japanese Neural Network Society 2007 Young Researcher Award**

Grant and Stipend

- 2010-2011 **Excellent Young Researchers Overseas Visit Program by JSPS**, Stipend
- 2008-2011 **Research Fellowship for Young Scientists by JSPS***, Stipend and Research Grant
- 2006-2008 **Research Fellowship for Young Scientists by JSPS***, Stipend and Research Grant
- 2004-2006 **Japan Student Services Organization**, Stipend
- 2002-2004 **Johns Hopkins University, Dept. of Neuroscience (Solo Snyder Foundation)**, Tuition and Stipend
- 2000-2002 **Murata Overseas Scholarship**, Tuition, Stipend, and Research Grant

* JSPS: Japan Society for the Promotion of Science

Letter

- **Letter of Appreciation from the RIKEN president Ryoji Noyori (2009/10)**

Travel Award

- 2006/12 **The 21st Century COE Center for Diversity and Universality in Physics**. NIPS2006 Travel Aid
- 2006/5 **Tamura Memorial Foundation, Frontiers in Dynamics: Physical and Biological Systems**. Travel Aid
- 2006/1 **Japanese Neural Network Society**. Travel Aid

| Research | Publications | Conferences | Awards | Curriculum Vitae |

Curriculum Vitae

[Please click here for my full CV.](#)